

There's text data everywhere now — tech docs, science papers, social media posts, customer reviews. Companies are figuring out this stuff contains gold but getting useful insights from it is tough. Old-school keyword searching just doesn't cut it anymore. Think about it — you search for 'car issues' and miss all the posts about 'automobile problems'. Or someone complains about their 'screen freezing' but your system is looking for 'system crashes'.

That's why text embeddings have become such a big deal. They turn words into number vectors that supposedly capture meaning. Big tech has thrown billions at this — OpenAI, Meta's RoBERTa, Google's BERT, plus tons of open source options. These are the engines running modern NLP systems.

But here's the weird part — we don't really understand how these models work in real life. This leads to:

- Expensive mistakes when systems don't deliver what businesses need.
- Frustrating user experiences when searches miss relevant content.
- Performance that varies wildly across user groups, content types, and languages.
- Wasting resources on overcomplicated models.

Why this matters for real businesses

This stuff directly impacts many industries.

Manufacturing:

- Finding parts and components despite varying naming conventions.
- Identifying similar production issues across different factory reports.
- Analysing maintenance logs to predict equipment failures.

Education:

- Retrieving learning resources that match student queries regardless of vocabulary level.
- Understanding student feedback across different age groups and language abilities.

- Connecting related concepts across different subjects and curricula.

Legal:

- Identifying relevant case law despite variations in legal terminology.
- Detecting contractual similarities despite different phrasing.
- Improving document discovery across multiple jurisdictions and legal traditions.

Knowing how your embedding model finds similar results helps you pick the right model, figure out preprocessing needs, and improve your systems. The end result? More reliable, efficient, and fair language processing.

Text embeddings in plain language

Think of text embeddings like translating words into a secret code of numbers. Each word gets turned into a long string of numbers (like 300 or more). The magic happens in how these numbers relate to each other — words with similar meanings end up with similar number patterns.

I use these every day without thinking about it:

- When my music app suggests songs based on my listening habits
- When my phone predicts what I'm about to type next
- When I search for 'good dog food' and it shows results for 'quality canine nutrition' too.

These number translations work behind the scenes in:

- Smart speakers understanding what you're asking for
- Language translation apps
- Email spam filters
- News article categorisation
- Social media content moderation

The success of these tools comes down to how well they handle our messy human language. If a system can't tell that "sooooo amazing!!!" and "really good" are positive reactions or think "not working" and "working perfectly" are basically the same thing, we've got problems. And trust me, we've got problems.

My wake-up call with embeddings

It hit me during a late-night testing session. I was building a document system for an engineering firm and needed to find specifications for 'materials without asbestos'. The system confidently returned documents about asbestos-containing materials. I checked twice to make sure I hadn't messed up my query.

That moment changed everything for me. If these supposedly smart systems couldn't grasp something as simple as 'without', what other critical details were they missing? The benchmarks all looked impressive on paper, but what about when faced with how engineers, managers, and field workers actually communicate?

I spent the next 18 months putting every embedding model I could find through a gauntlet of real-world language tests. I built the tests based on actual user queries from various industries using the RAG Experiment Accelerator framework. The results weren't just disappointing — they revealed fundamental flaws in how these systems process meaning.

I'm writing this because I've watched brilliant teams invest thousands of hours building sophisticated AI systems, only to have them fail spectacularly when deployed. Not because the teams did anything wrong, but because nobody warned them about these invisible limitations.

Real stakes, real consequences

This isn't just academic nitpicking. The consequences are serious.

A healthcare company colleague told me their medical information retrieval system was missing relevant clinical docs because their embedding model couldn't handle abbreviations and terminology variations. In healthcare, missing documents can affect treatment decisions.

At a financial firm, their compliance monitoring missed policy violations because their embedding model couldn't recognise that passive voice ('funds were transferred') meant the same thing as active voice ('someone transferred funds').